Table summarizing the observed statistics for each benchmark program:

Benchmark	**Instructions Executed**	**Cycles Taken**	**CPI**	**OF Stage Stalls (Number of times)**	**OF Stage Stalls (Number of Cycles)**	**Wrong Branch Path Instructions**
Descending	279	548	1.96	53	91	176
Even or Odd	8	14	1.75	3	4	0
Fibonacci	80	137	1.71	12	23	32
Palindrome	51	100	1.96	18	33	14
Prime	31	51	1.65	5	8	10

The benchmark programs show variation in performance based on instruction count, data dependencies, and branch behavior. Programs like **Descending** and **Palindrome** exhibit higher CPI values due to a larger number of OF stage stalls caused by data hazards. In particular, Descending also shows a significant number of wrong branch executions because of frequent looping, indicating strong control dependencies. Similarly, Palindrome has a high CPI due to repeated use of registers, leading to increased data hazards.

On the other hand, simpler programs such as **Even or Odd** and **Prime** execute fewer instructions and therefore experience minimal OF stage stalls and fewer wrong branch path entries. This results in lower CPI values and better overall efficiency. The Fibonacci program, while executing a moderate number of instructions and encountering some stalls and branch mispredictions, still maintains a relatively low CPI, indicating balanced performance compared to more dependency-heavy programs.